

Two-Body Problem (Classical Orbit Analysis)

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This project performs a classical orbital analysis of an Earth-orbiting satellite using its position and velocity vectors at eight epochs. Orbital elements, ground-station range (ρ), elevation (σ) and azimuth (β) angles, and two-body propagation were calculated with an original MATLAB code that implemented vector algebra, coordinate transformations and numerical integration with ODE45.

The orbit was found to be elliptical with $a = 26565$ km, $e = 0.74$ and $i = 63.4^\circ$. Ground-station observations from the Boston Air Force Station resulted in $\rho = 39507$ km, $\sigma = 36.4^\circ$ and $\beta = 320^\circ$.

Propagation of the orbit confirmed conservation of specific energy and angular momentum and produced a 3-D orbit consistent with the calculated orbital elements.

I. Introduction & Literature Review

The practice of spacecraft mechanics and orbital dynamics is considered relatively modern, but, in reality, they are based on concepts long since justified by Johann Kepler and Isacc Newton [1]. Since then, methods have been developed to optimize/simplify the process of orbit analysis, bringing light to different numerical methods of viewing an orbit, many of which require complex mathematics that can be simplified through coding [2, 3, 4, 5].

In this project, a satellite's orbit about the Earth will be quantified and described through orbital elements, range and angle measurements relative to the New Boston Air Force Station (NBAFS), and the Two Body Problem using a position and velocity vector at a given epoch. All calculations and graphs are coded in MATLAB, allowing for consistent and accurate results.

The original method of visualizing an orbit is through Two Body Problem. This originates from Newton's law of universal gravitation, $\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} = 0$ (or more commonly shown, $F = G \frac{m_1 m_2}{r^2}$), stating that the gravitational influence of one object on another is inversely proportional to the square of the distance between them, and proportional to the mass of the objects [1]. In Earth-satellite relationships, it is assumed that the two bodies are point masses, there are no outside forces acting on the system, and the mass of the satellite is negligible compared to the mass of Earth [6,7]. Using this equation along with an initial position and velocity vector, the method of propagation can

be utilized to come up with a position and velocity vector at any given time. Plotting position produces a 3-dimensional graph of the orbit. All vectors are set on the standard ECI coordinate frame ($\hat{I}, \hat{J}, \hat{K}$), fixed at the Earth's center but non-rotational, where $\hat{I}$ points to the vernal equinox and $\hat{K}$ points to the North Pole. The $\hat{I} - \hat{J}$ plane is in Earth's equatorial plane [6, 7].

One limitation with the Two Body Problem stems from its output of six non-constant variables. While this method produces very reliable data, it becomes almost impossible to visualize without proper graphing and can become very tedious to calculate six variables for every individual time. By utilizing orbital elements, we can simplify the problem into five constant and one non-constant variable [1, 6, 7]. Fig. 1 in the Appendix illustrates all the orbital elements in reference to each other. The orbital elements consist of eccentricity (e), a measure of the orbit's shape, semimajor axis (a) which determines an orbits size. The three Euler angles: inclination (i), which describes the tilt of the orbital plane relative to the equator, argument of periapsis (ω), which describes the orientation of the orbit within its plane and right ascension of the ascending node (Ω) which describes the orientation of the ascending node within the equatorial plane; and the true anomaly (f) which determines the angular location of the satellite from the ascending node and is the only time-dependent variable. All orbital elements are also measured relative to the standard ECI frame [6, 7].

However, satellite's position in space cannot realistically be measured directly in the ECI coordinate frame from Earth's surface. Many ground stations instead use ground-angle and range measurements to characterize a satellite's orbit. This method measures the satellite from a frame SEZ fixed to the geographical location of a ground station, where $\hat{Z}$ points opposite to the direction of Earth's core, $\hat{S}$ points South, and $\hat{E}$ points East. The main measurements of this method are range (ρ), elevation angle (σ), azimuth angle (β), and their rates ($\dot{\rho}, \dot{\sigma}, \dot{\beta}$). These values can be converted to their corresponding position and velocity vectors in the ECI frame through ECI-SEZ coordinate transformations and vice versa [7, 8].

II. Theoretical Development

The first step is to compute the orbital elements (eccentricity, semi-major axis, inclination, true anomaly, argument of periapsis, and right ascending node) of a satellite orbiting the Earth using the given position ($\vec{r}_1 = [-15865\hat{I} + 13312\hat{J} + 41357\hat{K}] \text{ km}$) and velocity ($\vec{v}_1 =$

$[-0.9601\hat{i} - 1.1443\hat{j} + 0\hat{k}] \text{ km/s}$ vectors at an instantaneous point in time (t_1), and their magnitudes were calculated with the norm function in MATLAB. This function uses the formula,

$$\|\vec{r}\| = \sqrt{(r_x)^2 + (r_y)^2 + (r_z)^2} \quad (\text{Eq. 1}).$$

The eccentricity element of the orbit (e) is calculated from the following equation,

$$\vec{e} = \frac{1}{\mu_E} (\vec{v}_1 \times \vec{h}) - \hat{r}_1 \quad (\text{Eq. 2})$$

where $\vec{e}$ is the eccentricity vector, $\mu_E = 398600 \frac{\text{km}^3}{\text{s}^2}$ (gravitational constant of Earth), $\vec{h}$ is the angular momentum vector calculated by $\vec{h} = \vec{r} \times \vec{v}$ (Eq. 3), and $\hat{r}_1$ is the instantaneous position unit vector, $\hat{r}_1 = \frac{\vec{r}_1}{\|\vec{r}_1\|}$ (Eq. 4). This vector is normalized to find the magnitude of eccentricity. In the code, an IF statement was used to classify the conic type of the orbit based on the magnitude of eccentricity [9].

The semimajor axis is calculated using the specific orbital energy. Although the radial distance and velocity of our satellite will vary continuously throughout the orbit, the specific orbital energy will remain constant in an ideal two-body problem with no perturbations. Since gravity is a conservative source, any increase in potential energy will be balanced by a decrease in kinetic energy and vice versa. Therefore, the total energy/specific energy is given by:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \quad (\text{Eq. 5})$$

where r and v are the magnitudes of the position and velocity vectors and ε is the specific energy.

Once the specific energy is calculated, the semi-major axis (a), can be found using:

$$a = -\frac{\mu}{2\varepsilon} \quad (\text{Eq. 6})$$

The three Euler angles were calculated through the following equations:

$$i = \cos^{-1} \left(\frac{h_z}{\|\vec{h}\|} \right) \quad (\text{Eq. 7})$$

$$\Omega = \tan^{-1} \left(\frac{\hat{n}_y}{\hat{n}_x} \right) \quad (\text{Eq. 8})$$

$$\omega = \cos^{-1} \left(\frac{\vec{n} \cdot \vec{e}}{\|\vec{n}\| \|\vec{e}\|} \right) \quad (\text{Eq. 9})$$

Although in MATLAB, Eq. 8 was coded as $\Omega = \text{atan2}(n_y, n_x)$ so it can compute angles from vector components, void ambiguity and handle all quadrants correctly.

To calculate the argument of periapsis (ω), the node vector and its magnitude must first be

calculated using $\vec{n} = \vec{k} \times \vec{h}$ (Eq. 10), where $\vec{k}$ is taken as the vector $[0, 0, 1]$ in the ECI coordinate frame. Calculations of argument of periapsis require a quadrant check, so, if $\hat{e} \cdot \hat{k} < 0$, the equation will be modified to $\omega = 2\pi - \cos^{-1}\left(\frac{\vec{n} \cdot \vec{e}}{\|\vec{n}\| \|\vec{e}\|}\right)$ (Eq. 11). If this condition isn't met, Eq. 9 can be utilized.

True anomaly (f), which describes the location of the satellite in the orbit at an instant in time, was calculated in a similar way, using a quadrant check. If $\vec{r} \cdot \vec{v} \geq 0$:

$$f = \cos^{-1}\left(\frac{\vec{e} \cdot \vec{r}}{\|\vec{e}\| \|\vec{r}\|}\right) \quad (\text{Eq. 12})$$

else,

$$f = 2\pi - \cos^{-1}\left(\frac{\vec{e} \cdot \vec{r}}{\|\vec{e}\| \|\vec{r}\|}\right) \quad (\text{Eq. 13})$$

However, in MATLAB, a temporary value for ω and f were used to first calculate an angle, using Eq. 9 and 11, and then an IF statement was used to check the conditions and subtract by a full angular rotation.

After computing all the classical orbital elements (COEs), additional parameters were derived to further describe the satellite's position and motion along its orbit. The eccentric anomaly (E), which provides a geometric reference on the auxiliary circle; the mean anomaly (M), which represents the fraction of the orbital period elapsed since passing through the periapsis of the orbit; and the orbital period (T), calculated using Kepler's third Law of Planetary Motion. This law states that the square of the orbital period is proportional to the cube of the semi-major axis. The following formulas were used to obtain these quantities:

$$E = 2 \tan^{-1}\left(\sqrt{\frac{1-\vec{e}}{1+\vec{e}}} \tan \frac{f}{2}\right) \quad (\text{Eq. 14})$$

$$M = E - \vec{e} \sin(E) \quad (\text{Eq. 15})$$

$$T = 2\pi \sqrt{\frac{a^3}{\mu}} \quad (\text{Eq. 16})$$

After obtaining these quantities, the Keplerian orbit can be fully described. Having found E , M and T , the epoch time (time elapsed since the satellite passed through the periapsis, $(t_1 - t_p)$) was found by finding the mean motion (n), $n = \sqrt{\frac{\mu}{a^3}}$ (Eq. 18), which is the average angular speed of the body and utilizing the equation, $(t_1 - t_p) = \frac{M}{n}$ (Eq. 17). This determines the spacecraft's "clock angle" within the orbit.

Next, the effective potential (U_{eff}) was found in terms of position (r) using the following equation:

$$U_{eff}(r) = -\frac{\mu}{r} + \frac{h^2}{2r^2} \quad (\text{Eq. 19})$$

where (r) and (h) are magnitudes.

The radius of apoapsis (r_a) and radius of periapsis (r_p) were calculated for the limits of the effective potential graph using these equations: $r_a = a(1 + e)$ (Eq. 20); $r_p = a(1 - e)$ (Eq. 21).

The second method is used to find the station's range (ρ), elevation (σ) and azimuth (β) of the satellite from the NBAFS located at latitude $\Phi = 42.9^\circ$ and longitude $\lambda = 240.7^\circ$ during t_2 .

The code first converts the station's longitude and latitude to radians and constructs two rotation matrices to form the ECI to SEZ transformation,

$$R^T = R_2(90 - \Phi)R_3 = \begin{bmatrix} \sin\Phi & 0 & -\cos\Phi \\ 0 & 1 & 0 \\ \cos\Phi & 0 & \sin\Phi \end{bmatrix} \begin{bmatrix} \cos\lambda & \sin\lambda & 0 \\ -\sin\lambda & \cos\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{Eq. 22})$$

where λ and Φ are in radians. The rotational matrixes referenced are located in the Appendix.

The station's position vector in the ECI frame is calculated by,

$$\vec{r}_{site,ECI} = R^T * r_{site,SEZ} \quad (\text{Eq. 23})$$

Where $r_{site,SEZ}$ is the position of the station in the SEZ frame.

To find the topocentric vector in the ECI frame (ρ) we apply the following:

$$\vec{\rho}_{ECI} = \vec{r}_{sat,ECI} - \vec{r}_{site,ECI} \quad (\text{Eq. 24})$$

where $\vec{r}_{sat,ECI}$ is the position of the satellite at t_2 in the ECI frame. This is also converted to SEZ frame using the rotational matrix, $\vec{\rho}_{SEZ} = R^T * \vec{\rho}_{ECI}$ (Eq. 25).

The magnitude of $\vec{\rho}_{SEZ}$ was calculated using the norm to be $\rho=39507.239$ km and the elevation and azimuth were calculated using the following:

$$\sigma = \sin^{-1} \left(\frac{\rho_z}{\|\vec{\rho}\|} \right) = 36.38^\circ \quad (\text{Eq. 26})$$

$$\beta = 2\pi - \tan^{-1} \left(\frac{\rho_E}{-\rho_S} \right) = 320.02^\circ \quad (\text{Eq. 27})$$

Since the computed angle was negative, the 2π correction had to be added to the azimuth calculation due to quadrant correction.

The spacecraft's motion was then propagated in time, treating it as a two-body problem and following Newton's law of universal gravitation:

$$\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} = 0 \quad (\text{Eq. 28})$$

This equation was solved using MATLAB's function ODE45, which numerically integrates the two-body equations of motion using $\frac{d\vec{r}}{dt} = \vec{v}$ (Eq. 29) and $\frac{d\vec{v}}{dt} = -\frac{\mu}{r^3} \vec{r}$ (Eq. 30).

These equations describe the time evolution of the satellite's position and velocity under the influence of Earth's gravitational field and are expressed as a system of first-order differential equations. ODE45 adjusts the integration step size to maintain accuracy while optimizing computational efficiency. By providing the derivative functions (Eq. 29 and 30), initial conditions (position and velocity vectors at t_1), and the desired time span, ODE45 calculates the satellite's trajectory over multiple intervals [10]. Eight equal intervals from t_1 to $t_8 = t_1 + T$ were used to calculate the orbital elements at each epoch, and 1000 equal intervals from t_1 to $t_{1000} = t_1 + 2T$ to graph the 3-dimensional orbital trajectory of the satellite.

III. Results & Discussion

After calculating the orbital elements at t_1 using $\vec{r}_1$ & $\vec{v}_1$, the semimajor axis of the orbit is found to be $\sim 26600 \text{ km}$, the eccentricity is 0.741, inclination is 63.4° , the right ascending node is 50.0° , argument of periapsis is 270° , and the true anomaly is 180° . The angular momentum is calculated as $69089 \frac{\text{km}^2}{\text{s}}$, and the specific orbital energy is $-7.502275 \frac{\text{km}^2}{\text{s}^2}$. Observing that $0 < e < 1$ and referencing Table 1 in the Appendix, the orbit conic type can be defined as elliptical.

Fig. 2 shows the effective potential graph that was plotted using the calculated angular momentum at t_1 . The produced equation for said graph is $U_{eff}(r) = -\frac{398600}{r} + \frac{2.387 \times 10^9}{r^2}$. The blue line represents the change in effective potential energy in terms of the radius of the orbit, the dotted black lines marking the limits of the orbit (perigee and apogee). The solid red line marks the specific energy calculated for t_1 .

Given the time at periapsis to be zero, the time after periapsis to t_1 was found to be 21545 sec. Propagating this orbit (starting at t_1 and continuing for a full period) for 8 intervals and solving for the orbital element at each epoch produces Table 2. The results for each element are constant for each interval except for true anomaly, which varies over time. The specific energies and angular momentums were also calculated at each interval and plotted over the interval time, as shown in Fig. 5 a & b. With insignificant variations, the graphs show specific energy and angular

momentum to be constant throughout the eight epochs.

From the NBAFS, the ground-angles and range were found to be $\rho = 39500 \text{ km}$, $\sigma = 36.4^\circ$, and $\beta = 320^\circ$.

Lastly, the orbit was propagated using the Two Body Problem from t_1 to $t_1 + 2T$ (propagated over two orbital periods) with 1000 intervals. The calculated position vectors were plotted to create a 3-dimensional model of the orbit (Fig. 3). The model confirms the orbit as elliptical and shows t_1 at the apoapsis ($f_1 = 180^\circ$).

The specific energy and angular momentum were also calculated over 1000 intervals and graphed over time (Fig. 5 c & d). The graphs produced are nearly identical to Fig. 5 a & b, confirming that specific energy and angular momentum are constant throughout orbit.

IV. Conclusions & Future Work

This project focused on conducting classical orbital analysis on an undefined orbit using an instantaneous position and velocity from epoch t_1 . The orbit was defined through three different methods: orbital elements, ground-angle and range, and the two-body problem. Each was utilized in an original MATLAB code, successfully producing the orbital elements; azimuth, range, and elevation; and 3-dimensional orbit trajectory graph.

Future improvements to this could include expanding the types of orbits it can handle. Currently, the code assumes an elliptical orbit, with formulas for the eccentric anomaly (E) and mean anomaly (M) valid only for $0 < e < 1$. Additionally, equations could be implemented for hyperbolic orbits (using the hyperbolic anomaly, H) circular or parabolic orbits, and the effective potential (U_{eff}) plot could be updated to account for negative semi-major axes in hyperbolic trajectories. Special cases where some orbital elements are undefined should also be considered [5,6].

To enhance usability, a user-input feature could allow the code to work for any position and velocity vector. The code could also be optimized to reduce the number of lines and redundant calculations, improving overall efficiency. Finally, adopting canonical units would further simplify calculations, reduce numerical errors and make the code more general for other two-body problems.

V. References

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VI. Appendix

Table 1. Properties of Conics

Energy	Semi-Major Axis	Eccentricity	Conic Section
$\varepsilon = \varepsilon_{min} = -\frac{\mu^2}{2h^2}$	$a = \frac{h^2}{\mu}$	$e = 0$	Circle
$\varepsilon_{min} < \varepsilon < 0$	$a > 0$	$0 < e < 1$	Ellipse
$\varepsilon = 0$	$a = \infty$ (<i>undefined</i>)	$e = 1$	Parabola
$\varepsilon > 0$	$a < 0$	$e > 1$	Hyperbola

Table 2. Orbital Elements at all time periods t1 to t8

	a (km)	e	i (deg)	Ω (deg)	ω	f
t1	26565	0.741095	63.4	50.0024	270	180
t2	26565	0.741095	63.4	50.0024	270	191.85
t3	26565	0.741095	63.4	50.0024	270	207.36
t4	26565	0.741095	63.4	50.0024	270	243.54
t5	26565	0.741095	63.4	50.0024	270	116.46
t6	26565	0.741095	63.4	50.0024	270	152.64
t7	26565	0.741095	63.4	50.0024	270	168.15
t8	26565	0.741095	63.4	50.0024	270	180

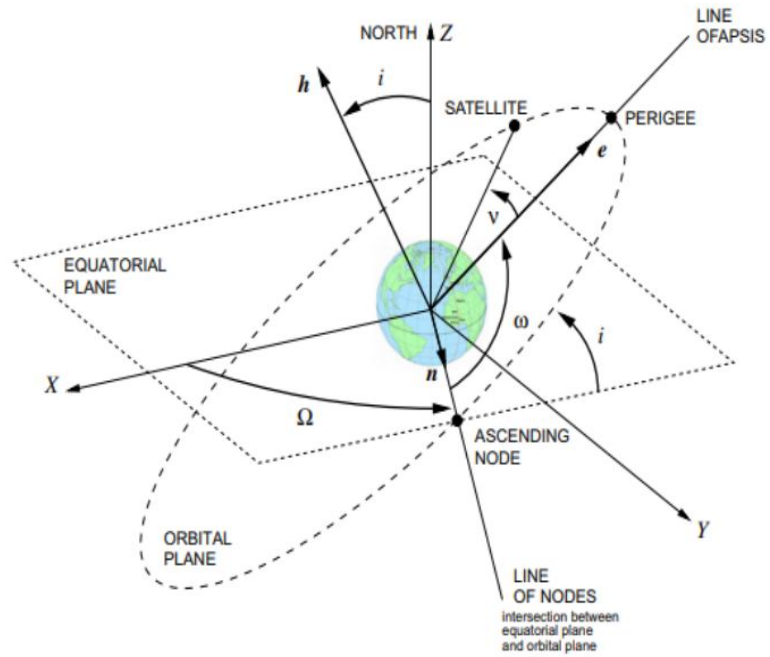


Fig. 1. OEs diagram [5]

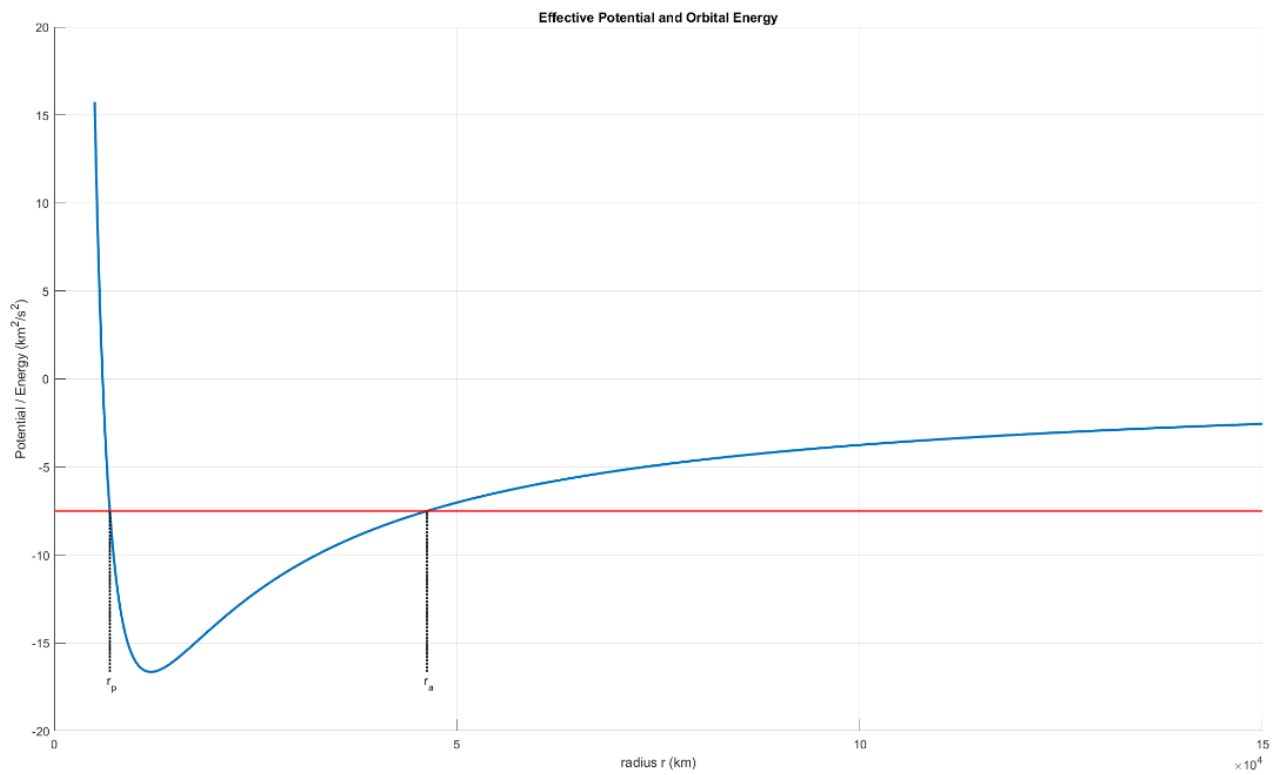


Fig 2. Effective Potential Energy vs Radius Plot at time t_1

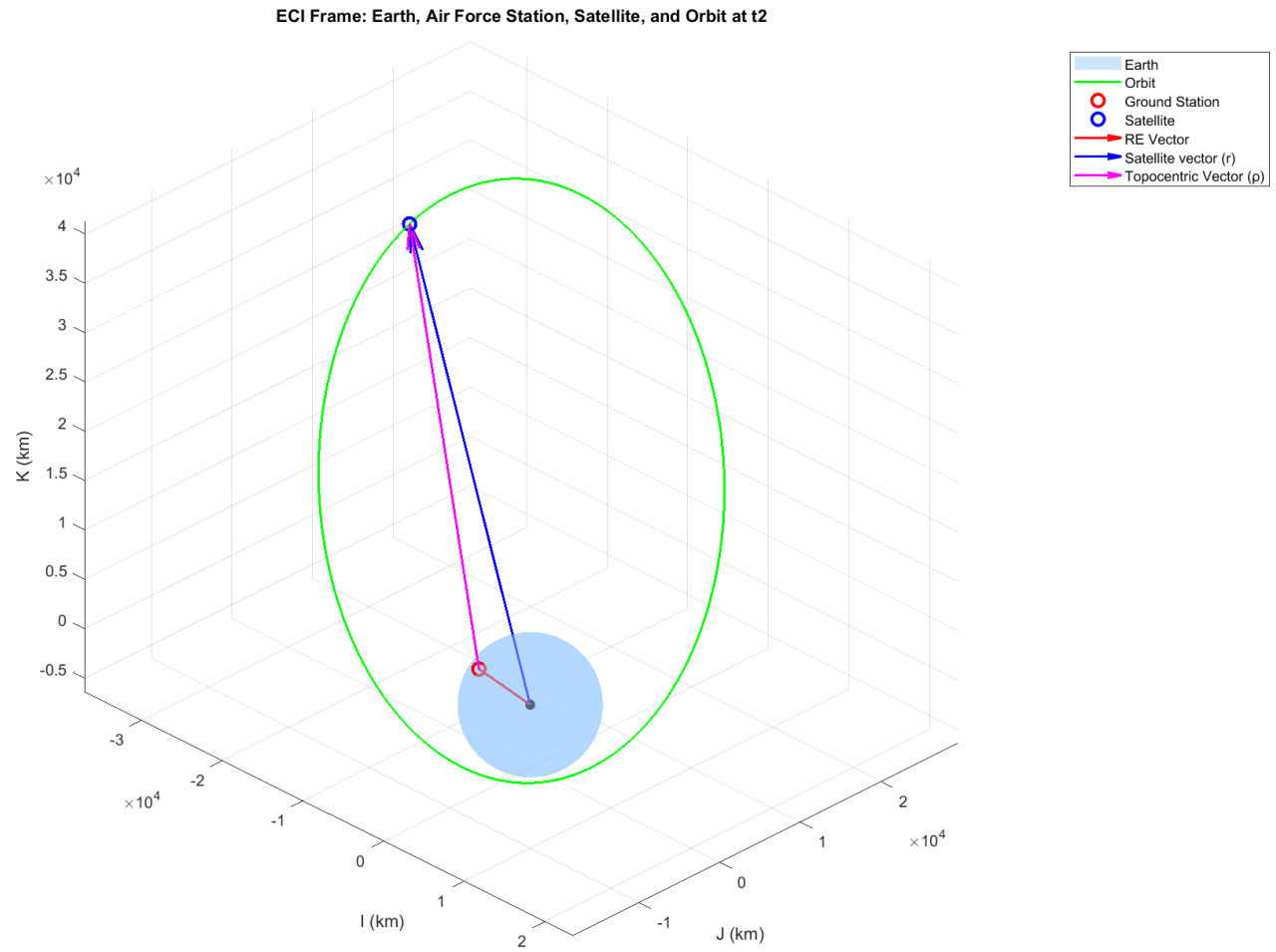


Fig 3. 3D range, elevation and azimuth for Boston Air Force

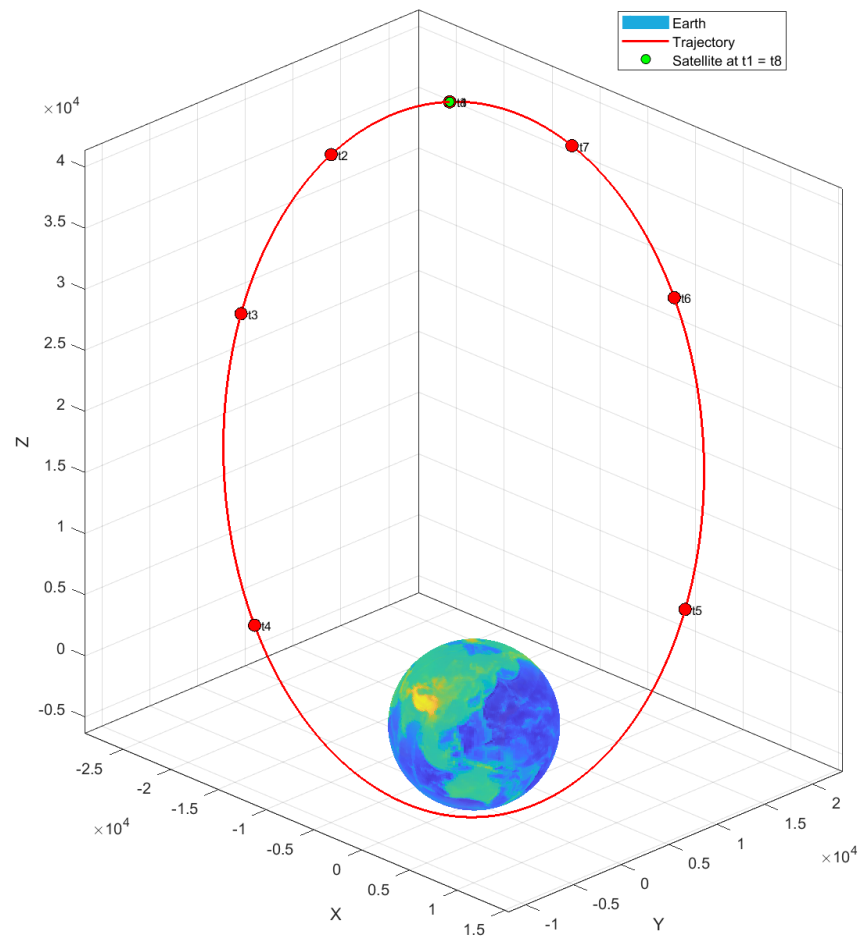


Fig 4. 3D Orbit plot

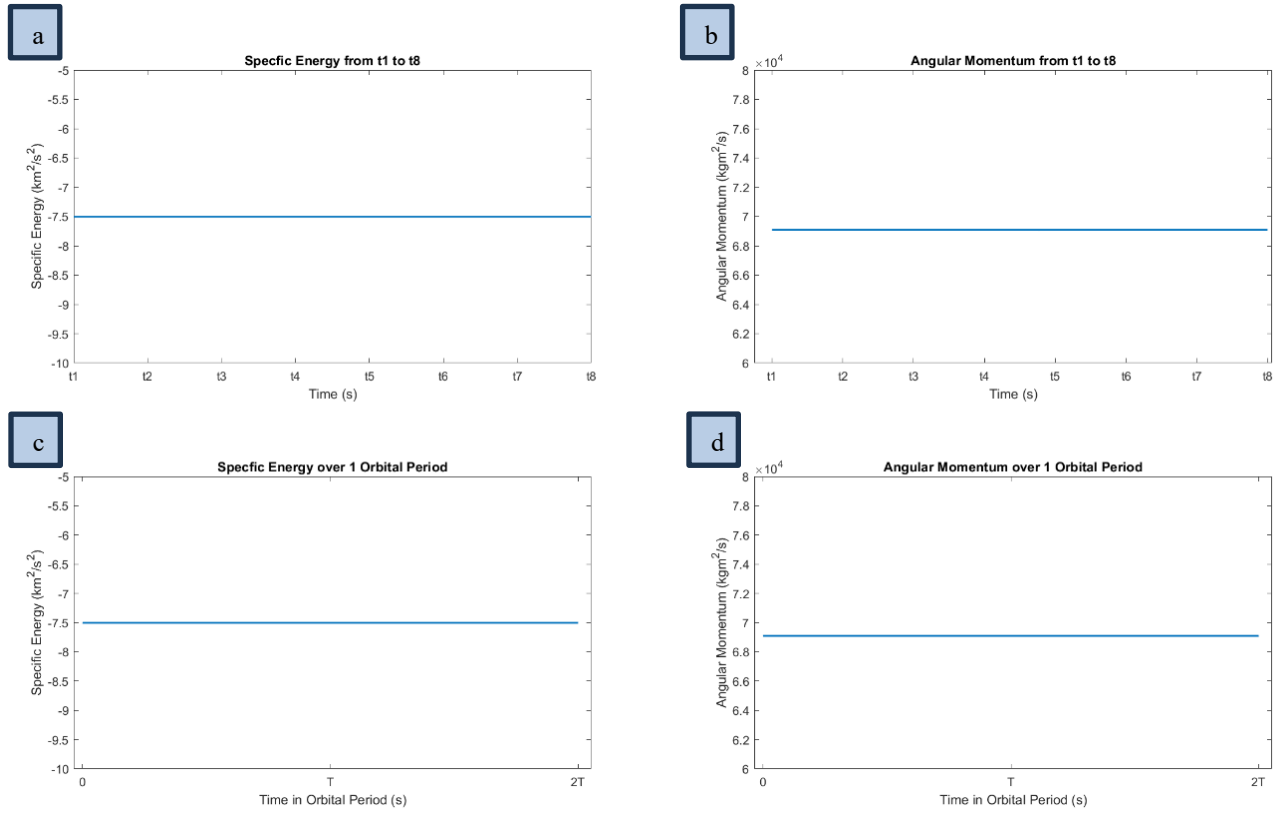


Fig 5. Variation of Specific Energy and angular momentum over two time periods

Extended Calculation of range, elevation and azimuth:

$$R_3(\lambda) = \begin{bmatrix} \cos\lambda & \sin\lambda & 0 \\ -\sin\lambda & \cos\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2(90 - \Phi) = \begin{bmatrix} \sin\Phi & 0 & -\cos\Phi \\ 0 & 1 & 0 \\ \cos\Phi & 0 & \sin\Phi \end{bmatrix}$$

$$R^T = R_2(90 - \Phi)R_3 = \begin{bmatrix} \sin\Phi & 0 & -\cos\Phi \\ 0 & 1 & 0 \\ \cos\Phi & 0 & \sin\Phi \end{bmatrix} \begin{bmatrix} \cos\lambda & \sin\lambda & 0 \\ -\sin\lambda & \cos\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Where $\lambda = 240.7^\circ = 4.20 \text{ rad}$ and $\Phi = 42.9^\circ = 0.748 \text{ rad}$

The station's SEZ vector is calculated using the following:

$$\vec{r}_{site,SEZ} = \begin{bmatrix} 0 \\ 0 \\ R_E + H \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 6378.137 \end{bmatrix} \text{ km}$$

Using the rotation matrix:

$$\vec{r}_{site,ECI} = R^T * \vec{r}_{site,SEZ} = \begin{bmatrix} -2287 \\ -4075 \\ 4342 \end{bmatrix} \text{ km}$$

Then, for the satellite, $\vec{r}_{sat,ECI}$ will be the position vector at t2. To find the topocentric vector in the ECI frame (ρ) we apply the following:

$$\vec{r}_{sat,ECI} = \begin{bmatrix} -20387.55 \\ 5422.94 \\ 38149.61 \end{bmatrix} \text{ km}$$

$$\vec{\rho}_{ECI} = \vec{r}_{sat,ECI} - \vec{r}_{site,ECI} = \begin{bmatrix} -18101 \\ 9497.5 \\ 3380.8 \end{bmatrix} \text{ km}$$

Converting it to SEZ frame:

$$\vec{\rho}_{SEZ} = R^T * \vec{\rho}_{ECI} = \begin{bmatrix} \rho_S \\ \rho_E \\ \rho_Z \end{bmatrix} = \begin{bmatrix} -24374 \\ -20433 \\ 23436 \end{bmatrix} \text{ (km)}$$

$$\rho = \|\rho\| = \sqrt{(-24374)^2 + (-20433)^2 + (23436)^2} = 39507.239 \text{ km}$$

$$\sigma = \sin^{-1}\left(\frac{\rho_Z}{\|\rho\|}\right) = 36.38^\circ$$

$$\beta = 2\pi - \tan^{-1}\left(\frac{\rho_E}{-\rho_S}\right) = 320.02^\circ$$